Comprehensive PHP MySQLi CRUD Reference Blueprint

High-visibility light mode edition with large, clear fonts for direct implementation

1. Permanent Database Connection Template

```php
<?php
// 1. Default Server Configuration
$host    = "localhost";
$user    = "root";
$pass    = "";
$db_name = "my_database";

// 2. Open Connection via mysqli Object
$conn = new mysqli($host, $user, $pass, $db_name);

// 3. Validate Connection Status
if ($conn->connect_error) {
    die("Database Connection Failed: " . $conn->connect_error);
}

// 4. Force Unified UTF-8 Encoding
$conn->set_charset("utf8mb4");
?>
```

2. Create / Insert Record Template

```php
<?php
// 1. Sanitize and Clean Incoming User Inputs
$val1 = htmlspecialchars(strip_tags($_POST['input_str']));
$val2 = intval($_POST['input_num']);

// 2. Prepare Secure SQL Statement
$sql = $conn->prepare("INSERT INTO `table_name` (`col_str`,
`col_int`) VALUES (?, ?)");

// 3. Bind Parameters (s for string, i for integer)
$sql->bind_param("si", $val1, $val2);

// 4. Execute and Check Result
if ($sql->execute()) {
    echo "Record Inserted Successfully";
} else {
    echo "Insertion Error: " . $sql->error;
}

// 5. Close Statement to Free Resources
$sql->close();
?>
```

3. Read / Select Records Template

```php
<?php
// 1. Prepare Select Query Statement
$sql = $conn->prepare("SELECT `col_str`, `col_int` FROM
`table_name`");
$sql->execute();

// 2. Fetch and Store Result Set
$result = $sql->get_result();

// 3. Validate Row Existence and Loop Through Data
if ($result->num_rows > 0) {
    while ($row = $result->fetch_assoc()) {
        echo "Text: " . $row['col_str'] . " | Number: " .
$row['col_int'] . "<br>";
    }
} else {
    echo "No Records Found";
}

// 4. Close Statement
$sql->close();
?>
```

4. Update / Modify Records Template

```php
<?php
// 1. Collect Target Data and Identifier
$new_val = htmlspecialchars(strip_tags($_POST['new_value']));
$id      = intval($_POST['id']);

// 2. Prepare Conditioned Update Statement
$sql = $conn->prepare("UPDATE `table_name` SET `col_str` = ? WHERE
`id` = ?");

// 3. Bind Updated Payload to the Specific Identifier
$sql->bind_param("si", $new_val, $id);

// 4. Execute and Validate Action
if ($sql->execute()) {
    echo "Record Updated Successfully";
} else {
    echo "Update Error: " . $sql->error;
}

$sql->close();
?>
```

5. Final Delete Records Template

```php
<?php
// 1. Secure and Cast the Target Identifier
$id = intval($_POST['id']);

// 2. Prepare Exact Delete Statement Based on Primary Identifier
$sql = $conn->prepare("DELETE FROM `table_name` WHERE `id` = ?");

// 3. Bind Identifier Parameter as Integer
$sql->bind_param("i", $id);

// 4. Execute and Perform Secure Clean Close
if ($sql->execute()) {
    echo "Record Deleted Successfully";
} else {
    echo "Deletion Error: " . $sql->error;
}
$sql->close();
$conn->close(); // Safely terminate database connection
?>
```

💡 Developer's Data Binding Reference Guide for bind_param():

Ensure letter order exactly matches your statement's question marks (?):
`i` for Integers (Whole Numbers) | `s` for Strings (Text / Dates) | `d` for Doubles (Floating Point Decimal Numbers) | `b` for Blobs (Binary Data / Files / Images).